On the Fragility of Multi-Task U-Nets in Computational Pathology: Empirical Replication, Metric Artifacts, and Task Interference

Muhammad Abdullah Ali[1⋆], Muhammad Ibrahim Kiani[1], and Muhammad Abdullah Aamir[1]

Department of Data Science, National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan
{i232523, i232536, i232538}@isb.nu.edu.pk

Abstract. Simultaneous classification and segmentation via hard-parameter-sharing U-Nets represents an appealing paradigm for computational pathology. Recently, high classification accuracies ($86\%-90\%$) and near-perfect segmentation Dice scores ($95\%-99\%$) were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. In this work, we conduct a comprehensive, 26-experiment replication and methodological audit across four medical imaging datasets: TCGA-LGG brain-tumor MRI, PANDA Prostate Biopsies, SIIM-ACR Pneumothorax chest radiographs, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (`GroupKFold`) and standard non-empty foreground Dice evaluation, we demonstrate that the published claims are methodologically incompatible with standard metric formulations and clinically unfeasible under patient isolation. Specifically, on fine-grained 6-class prostate cancer grading (PANDA), empirical top-1 accuracy spans $30.89\%-46.15\%$ and Dice spans $28.94\%-44.34\%$ across all 12 evaluated PANDA configurations, with the unweighted 10^{-3} baselines (Runs 03–04) achieving $45.15\% - 46.15\%$ / $43.30\% - 44.34\%$ (vs 88.0% and 99.0% claimed). We identify three systemic root causes: (1) background-Dice inflation artifacts on empty masks, (2) patient-level data leakage, and (3) fragility of fine-grained tasks to the joint optimization recipe (learning rate $\times$ dynamic gradient balancing $\times$ stain normalization), which our matched $2\times2\times2$ campaign partially decomposes. Furthermore, we evaluate dynamic gradient balancing (GradNorm), color deconvolution, and skip-connection ablations with strictly matched configurations: the skip-connection ablation is a verified null result (both coarse macro-lesions and fine micro-glandular structures are robust to the bottleneck-only representation in this architecture), whereas removing Macenko stain normalization significantly *improves* fine-grained classification. We provide a rigorous methodological framework and benchmark to prevent metric artifacts in future multi-task pathology studies.

Keywords: Computational Pathology · Multi-Task Learning · U-Net · Reproducibility · Metric Artifacts · GradNorm.

⋆ Corresponding author: i232523@isb.nu.edu.pk

1 Introduction

Deep learning systems in digital pathology and radiology are increasingly tasked with multi-faceted clinical reasoning. Simultaneous prediction of global slide- or tile-level diagnostic categories (classification) alongside precise lesion delineation (segmentation) is clinically desirable [5,3]. Hard parameter sharing via an encoder–decoder backbone (e.g., U-Net [9] with dual prediction heads) offers computational efficiency and the theoretical promise of regularized, synergistic latent representations.

Recently, Rhanoui et al. [8] claimed that a standard U-Net with lightweight encoders (VGG16 [11] and MobileNetV2 [10]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting $86\% - 90\%$ classification accuracy and $95\% - 99\%$ Dice similarity coefficients.

In this paper, we report the results of an exhaustive 26-run replication campaign and methodological teardown. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (`GroupKFold`), multi-class grading on PANDA prostate biopsies collapses from the claimed 88.0% accuracy / 99.0% Dice down to an empirical range of $30.89\% - 46.15\%$ Acc / $28.94\% - 44.34\%$ Dice across all 12 configurations, with the unweighted 10^{-3} baselines (Runs 03–04) achieving $45.15\% - 46.15\%$ / $43.30\% - 44.34\%$.
2. **The Background-Dice Metric Artifact:** Near-perfect (99%) Dice metrics in small-lesion and sparse-mask datasets (such as SIIM-ACR pneumothorax) are artifacts of evaluating Dice over true-negative background slices ($|Y| = |\hat{Y}| = 0$) or including background pixels, which trivially yields Dice $\equiv 1.0$.
3. **Controlled Multi-Task & Ablation Analysis:** Through strictly matched controls (learning-rate pairs, the static loss-ratio sweep $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$, GradNorm [4] on/off pairs, Macenko on/off pairs, and skip-connection ablations), we identify which factors of the optimization recipe genuinely move fine-grained performance and where matched effects are null (e.g., skip connections in this architecture); we also document a phase-gating defect that silently converted one intended GradNorm arm into a static control.

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 26 runs, evaluation checkpoints, and generated figures are publicly available at `https://github.com/ApatheticMioz/cancer-pathology-dl` [1].

2 Multi-Task Formulation & Evaluation Metrics

2.1 Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input pathology image tile. A shared convolutional backbone $f_\theta(\mathbf{x})$ produces a latent representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The classification branch applies global average pooling followed by a linear projection:

$$\hat{\mathbf{y}}_{cls} = \mathrm{Softmax}(\mathbf{W}_{cls} \cdot \mathrm{GAP}(\mathbf{z}) + \mathbf{b}_{cls}) \in \mathbb{R}^K, \tag{1}$$

where K is the number of diagnostic classes. The segmentation branch passes $\mathbf{z}$ through a sequence of transposed convolutions with skip connections from encoder stages:

$$\hat{\mathbf{y}}_{seg} = \sigma(\mathrm{Dec}_\phi(\mathbf{z}, \{\mathbf{s}_l\}_{l=1}^L)) \in [0,1]^{H\times W\times M}, \tag{2}$$

where M denotes the number of segmentation target channels and $\sigma(\cdot)$ is the sigmoid activation.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls}\mathcal{L}_{CE}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \hat{\mathbf{y}}_{seg}), \tag{3}$$

where $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$.

2.2 The Empty-Mask Dice Metric Artifact

The standard Sørensen–Dice coefficient between binary ground truth $Y \in \{0,1\}^{H\times W}$ and prediction $\hat{Y} \in \{0,1\}^{H\times W}$ is:

$$\mathrm{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}. \tag{4}$$

A critical methodological divergence occurs when an image tile contains no target lesion ($|Y| = 0 \wedge |\hat{Y}| = 0$). In standard biomedical benchmarking (e.g., Metrics Reloaded [7]), two conventions exist:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($|Y| > 0$).
- **Zero-Score Penalty:** Assign Dice = 1.0 if $|Y| = 0 \wedge |\hat{Y}| = 0$, but Dice = 0.0 if $|\hat{Y}| > 0$.

When a model naively predicts near-zero masks on a dataset with $60\% - 80\%$ negative slices, assigning Dice = 1.0 artificially inflates the aggregate mean Dice to $> 95\%$, masking total segmentation collapse.

2.3 Adaptive Gradient Balancing via GradNorm

To decouple the empirical interaction between λ_{seg} and λ_{cls}, the v2 phase employs a parameterized variant of GradNorm [4]. Let W denote *all* trainable parameters of the shared U-Net encoder (i.e., every convolutional and normalization block of the backbone, not only its final block), and let

$$G_W^{(i)}(t) = \|\nabla_W \mathcal{L}_i(t)\|_2 \tag{5}$$

be the gradient norm of task $i \in \{seg, cls\}$ with respect to W at minibatch t. The balancer maintains learnable task weights $w_i(t)$ — initialized at $w_{seg}(0) = 5.0$ and $w_{cls}(0) = 1.0$, inherited from the static configuration, and held as learnable log-weights — and, once per minibatch, minimizes

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg,cls\}} \left| w_i(t)\, G_W^{(i)}(t) \;-\; \bar{G}_W(t) \cdot [r_i(t)]^{\alpha} \right|, \tag{6}$$

where $r_i(t) = \dfrac{\mathcal{L}_i(t)/\mathcal{L}_i(0)}{\frac{1}{2}\sum_{j \in \{seg,cls\}} \mathcal{L}_j(t)/\mathcal{L}_j(0)}$ is the relative loss ratio (mean-normalized across tasks, $\mathcal{L}_i(0)$ being the loss recorded at the first minibatch), $\bar{G}_W(t)$ is the mean gradient norm across tasks, and $\alpha = 1.5$ is fixed for all v2 runs. After every optimizer step, the weights are renormalized to $\sum_i w_i(t) = T = 2$.

We disclose two deviations from the reference formulation for reproducibility. First, the reference GradNorm applies a negative exponent to the normalized training rate ($[r_i]^{-\alpha}$ in the notation of Chen et al.); our implementation uses the positive-exponent convention $\bar{G}_W[r_i]^{+\alpha}$ on the mean-normalized loss ratio. Second, whereas the reference optimizes the balancing weights with a dedicated inner-loop optimizer, our log-weights are updated by the *same* `Adam` optimizer as the network (no inner loop), once per minibatch. All v2 runs share this identical variant and the same α, so every within-group (v2-vs-v2) comparison in this paper is a comparison within one and the same balancing implementation.

3 The 26-Run Empirical Reproduction Campaign

3.1 Datasets and Experimental Protocol

We evaluated four benchmark datasets encompassing digital pathology and medical imaging:

1. **TCGA-LGG Brain Tumor:** 2-class (lesion vs. normal) slice classification and FLAIR-abnormality segmentation on 2D 256×256 three-sequence MRI slices ($N = 3,929$ slices, 110 patients; the 2D MRI release of the TCGA lower-grade glioma collection).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading (ISUP 0–5) and epithelial gland segmentation ($N = 10,616$ tiles, with patient-level biopsy isolation via `GroupKFold`).
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 12,047$ images).
4. **PanNuke:** External 19-tissue multi-organ histology benchmark across 5 cell-nuclei classes ($N = 7,901$ patches). No multi-task claims were published for this benchmark, so it serves as an unclaimed external control.

All models were trained with PyTorch using fixed random seeds and the `Adam` optimizer with a constant per-phase learning rate and default (zero) weight decay; no learning-rate schedule (e.g., cosine annealing) or additional regularization is applied anywhere in the released codebase. In Phase v1 (Runs 01–06), models replicate the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), models incorporate dynamic GradNorm ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [6]. Groups 3–5 (Runs 13–26) systematically explore multi-organ generalizability, hyperparameter isolation, and architectural skip ablations. Input preprocessing note: all v1 runs use raw inputs by construction; among the v2 runs, Runs 17 and 23–24 are the only raw-input exceptions (marked $\diamondsuit$ in Table 1). Three additional matched control runs (27–29) that complete the Group-4 factorial design are defined in the released pipeline (see §4.3 for details) and are pending at the time of submission. All 26 runs are tabulated in Table 1 (Section 4).

4 Discussion & Methodological Anatomy

4.1 The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 12 PANDA configurations tested, empirical top-1 accuracy spans $30.89\% - 46.15\%$ and Dice spans $28.94\% - 44.34\%$, with the unweighted 10^{-3} baselines (Runs 03–04) achieving 45.15% / 43.30% (VGG16) and 46.15% / 44.34% (MobileNetV2) — the best reproducible performance on this task in our campaign.

This outcome is fully consistent with established clinical and machine learning benchmarks. In the international PANDA challenge ($> 1,000$ participating teams) [2], top-performing multi-instance learning (MIL) algorithms on whole-slide images achieved quadratic weighted kappa scores of 0.86–0.90, corresponding to tile-level top-1 multi-class accuracies in the $40\% - 52\%$ range. Human pathologists achieve inter-observer agreement kappa of only 0.65–0.75 on biopsy cores due to continuous histological transitions between Gleason patterns 3, 4, and 5. Claiming 88.0% accuracy on 128×128 isolated patches with a basic 2D CNN is unfeasible without patient-level data leakage (where tiles from the same biopsy core contaminate both train and test splits).

4.2 Controlled Skip Connection Ablation: A Verified Null Result

To evaluate the structural necessity of skip connections, we analyze strictly matched pairs (all other factors — learning rate, GradNorm, Macenko normalization — held identical):

- **Coarse macro-tumor structures (TCGA):** Comparing the skip-ablated MobileNetV2 (Run 25) against its matched baseline (Run 08; both at $\eta = 10^{-4}$, GradNorm, Macenko) shows accuracy shifting from 93.32% to 94.34% ($\Delta_{\text{Acc}} = +1.02$) while Dice remains essentially constant at 84.20% vs 84.12% ($\Delta_{\text{Dice}} = -0.08$).

Table 1: Comprehensive 26-run experimental results matrix across all five evaluation groups. V1 = original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw input); V2 = GradNorm variant ($\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization; see §2.3 for the exact variant). ✓ = GradNorm active; — = not used. Markers: × = skip connections zeroed (bottleneck-only decoding); ♢ = raw input without Macenko normalization (v1 runs are raw by construction and are unmarked). Rows 17–22 (Group 4) are raw-input PANDA × VGG16 controls; Run 18 was intended as the 10^{-3}-GradNorm arm but executed as a static 5:1 control (see §4.3). Paper Acc/Dice: published targets of Rhanoui et al. [8] (— = no published target). Acc Δ / Dice Δ: difference relative to those published targets. All entries in %.

# Configuration	Dataset	Encoder	LR η	GradNorm	Val Acc	Val Dice	Paper Acc	Paper Dice	Acc Δ	Dice Δ
Group 1: Baseline Reproductions (static 5:1, $\eta = 10^{-3}$, raw input)										
01 TCGA/vgg16 (Naked Baseline)	TCGA	VGG16	10^{-3}	—	88.82	76.02	89.0	97.0	−0.18	−20.98
02 TCGA/mobilenet_v2 (Naked Baseline)	TCGA	MobileNetV2	10^{-3}	—	94.60	87.56	90.0	98.0	+4.60	−10.44
03 PANDA/vgg16 (Naked Baseline)	PANDA	VGG16	10^{-3}	—	45.15	43.30	87.0	98.0	−41.85	−54.70
04 PANDA/mobilenet_v2 (Naked Baseline)	PANDA	MobileNetV2	10^{-3}	—	46.15	44.34	88.0	99.0	−41.85	−54.66
05 SIIM/vgg16 (Naked Baseline)	SIIM	VGG16	10^{-3}	—	83.19	77.74	82.0	99.0	+1.19	−21.26
06 SIIM/mobilenet_v2 (Naked Baseline)	SIIM	MobileNetV2	10^{-3}	—	83.47	77.74	87.0	99.0	−3.53	−21.26
Group 2: V2 Package (GradNorm $\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization)										
07 TCGA/vgg16 (GradNorm v2)	TCGA	VGG16	10^{-4}	✓	92.80	78.30	89.0	97.0	+3.80	−18.70
08 TCGA/mobilenet_v2 (GradNorm v2)	TCGA	MobileNetV2	10^{-4}	✓	93.32	84.20	90.0	98.0	+3.32	−13.80
09 PANDA/vgg16 (GradNorm v2)	PANDA	VGG16	10^{-4}	✓	30.89	31.59	87.0	98.0	−56.11	−66.41
10 PANDA/mobilenet_v2 (GradNorm v2)	PANDA	MobileNetV2	10^{-4}	✓	34.70	31.34	88.0	99.0	−53.30	−67.66
11 SIIM/vgg16 (GradNorm v2)	SIIM	VGG16	10^{-4}	✓	81.08	77.74	82.0	99.0	−0.92	−21.26
12 SIIM/mobilenet_v2 (GradNorm v2)	SIIM	MobileNetV2	10^{-4}	✓	83.00	77.74	87.0	99.0	−4.00	−21.26
Group 3: PanNuke 19-Tissue Multi-Organ Benchmark (external control; no published targets)										
13 PanNuke/vgg16 (Naked Baseline)	PanNuke	VGG16	10^{-3}	—	98.85	72.48	—	—	—	—
14 PanNuke/mobilenet_v2 (Naked Baseline)	PanNuke	MobileNetV2	10^{-3}	—	99.43	74.21	—	—	—	—
15 PanNuke/vgg16 (GradNorm v2)	PanNuke	VGG16	10^{-4}	✓	91.90	39.59	—	—	—	—
16 PanNuke/mobilenet_v2 (GradNorm v2)	PanNuke	MobileNetV2	10^{-4}	✓	96.68	60.54	—	—	—	—
Group 4: Optimization Teardown (PANDA × VGG16: LR, GradNorm, loss weighting; raw input)										
17 PANDA/vgg16 ♢ (Isolate LR: static $\eta = 10^{-4}$)	PANDA	VGG16	10^{-4}	—	43.35	37.99	87.0	98.0	−43.65	−60.01
18 PANDA/vgg16 (static 5:1; seed replicate of Run 20 — see §4.3)	PANDA	VGG16	10^{-3}	—	42.54	40.84	87.0	98.0	−44.46	−57.16
19 PANDA/vgg16 ($\lambda_{seg} : \lambda_{cls}$ = 1:1)	PANDA	VGG16	10^{-3}	—	42.73	41.72	87.0	98.0	−44.27	−56.28
20 PANDA/vgg16 ($\lambda_{seg} : \lambda_{cls}$ = 5:1, replicate)	PANDA	VGG16	10^{-3}	—	45.39	44.08	87.0	98.0	−41.61	−53.92
21 PANDA/vgg16 ($\lambda_{seg} : \lambda_{cls}$ = 1:10)	PANDA	VGG16	10^{-3}	—	41.83	38.06	87.0	98.0	−45.17	−59.94
22 PANDA/vgg16 ($\lambda_{seg} : \lambda_{cls}$ = 10:1)	PANDA	VGG16	10^{-3}	—	44.58	41.52	87.0	98.0	−42.42	−56.48
Group 5: Ablation Matrix (stain normalization ♢ and skip connections ×)										
23 PANDA/mobilenet_v2 ♢ (No-Macenko)	PANDA	MobileNetV2	10^{-4}	✓	40.21	31.57	88.0	99.0	−47.79	−67.43
24 PanNuke/mobilenet_v2 ♢ (No-Macenko)	PanNuke	MobileNetV2	10^{-4}	✓	99.36	73.33	—	—	—	—
25 TCGA/mobilenet_v2 × (No-Skips)	TCGA	MobileNetV2	10^{-4}	✓	94.34	84.12	90.0	98.0	+4.34	−13.88
26 PANDA/mobilenet_v2 × (No-Skips)	PANDA	MobileNetV2	10^{-4}	✓	35.31	28.94	88.0	99.0	−52.69	−70.06

Notes: ×: decoder skip connections zeroed (bottleneck-only representation). ♢: Macenko normalization disabled (raw input); v1 runs are raw by construction and unmarked. Acc Δ / Dice Δ are differences from the published reference targets of [8].

– **Fine micro-glandular structures (PANDA):** Comparing the matched MobileNetV2 configurations (Run 10: 34.70% Acc / 31.34% Dice vs Run 26, skip-ablated: 35.31% Acc / 28.94% Dice) yields $\Delta_{\text{Acc}} = +0.61$ and $\Delta_{\text{Dice}} = -2.40$.

In both cases the differences are small, and the 95% Wilson confidence intervals on top-1 accuracy overlap substantially (TCGA: $[91.57, 95.07]$ vs $[92.72, 95.96]$; PANDA: $[32.67, 36.73]$ vs $[33.27, 37.35]$). We therefore report these as *verified null results*: in this architecture, the bottleneck-only representation suffices for both coarse macro-lesion and fine micro-glandular segmentation, and zeroing the skip connections produces no detectable penalty in a single-run comparison. For transparency, we note that an earlier draft of this manuscript reported a 10.8 / 15.4-point "skip-connection paradox" on PANDA; that comparison (no-skip v2 vs the naked 10^{-3} v1 baseline) confounded the learning rate, dynamic balancing, and stain normalization, and we retract it. A directional claim would require multi-seed replication of the matched pairs.

Macenko stain-normalization ablation. The one matched ablation in this campaign that yields a statistically significant effect on classification is the Macenko normalization pair (Group 5, Runs 23–24): removing the normalization *increases* PANDA top-1 accuracy from 34.70% to 40.21% ($\Delta = +5.51$; 95% CIs $[32.67, 36.73]$ vs $[38.11, 42.31]$ are disjoint) and PanNuke accuracy from 96.68% to 99.36% ($\Delta = +2.68$; $[95.77, 97.59]$ vs $[98.96, 99.76]$ are disjoint), while Dice scores remain essentially unchanged (PANDA 31.34% $\rightarrow$ 31.57%; PanNuke 60.54% $\rightarrow$ 73.33%). On these fine-grained datasets, color deconvolution — a transformation designed to remove inter-dataset color variation — therefore measurably harms classification, consistent with over-smoothing of the chromatin/texture cues on which fine-grained grading relies. This result also partially decomposes the v1-vs-v2 PANDA gap of §4.1: part of that gap is attributable to Macenko itself, not to dynamic balancing or the learning rate.

4.3 Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA $\times$ VGG16) was designed as a $2 \times 2 \times 2$ factorial over the three factors of the v2 optimization package: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), dynamic gradient balancing (GradNorm on/off), and stain normalization (Macenko on/off). Two findings qualify the naive reading of this design.

Transparency note (Run 18). The intended 10^{-3}-GradNorm arm (Run 18) was launched with `-phase v1` on the assumption that GradNorm would activate by default. In fact, the v1 phase configuration hard-codes GradNorm off, and the released command line offered no override: Run 18 silently executed as a static 5:1 control — a seed replicate of Run 20 (accuracy 42.54% vs 45.39%, with overlapping 95% Wilson CIs $[40.43, 44.65]$ vs $[43.26, 47.52]$). We re-label it accordingly and exclude it from the GradNorm analysis. This incident is itself a methodological finding: an un-audited pipeline would have allowed a silently dropped experimental arm to masquerade as a treatment effect. The released code now exposes explicit `-enable-gradnorm`/`-disable-gradnorm` overrides, and every summary JSON and log records the *effective* GradNorm state of each run.

Valid matched subsets. With the design as executed, the following comparisons are defensible:

- **At $\eta = 10^{-4}$ (raw input, GradNorm off, Run 17 vs Run 09 with GradNorm on and Macenko on):** static 43.35% / 37.99% vs GradNorm 30.89% / 31.59% ($\Delta_{\text{Acc}} = -12.46$; 95% CIs $[41.23, 45.47]$ vs $[28.92, 32.86]$ are disjoint). The drop is real, but this pair simultaneously changes GradNorm *and* Macenko, so its attribution is confounded.
- **Macenko axis ($\eta = 10^{-4}$, GradNorm on, MobileNetV2; Run 10 vs Run 23):** 34.70% / 31.34% vs 40.21% / 31.57% ($\Delta_{\text{Acc}} = +5.51$; CIs disjoint) — the only factor in the v2 package that is cleanly isolated on PANDA, and it *helps* when removed (see §4.2).

– **At $\eta = 10^{-3}$ (static 5:1, raw; Runs 03/18/20):** 45.15%–45.39% — but the matching GradNorm-on cell was never executed (it was the lost Run-18 arm), so the GradNorm effect at the original learning rate is currently undetermined.

Because removing Macenko *improves* classification by +5.51 points (matched pair above), part of the 12.46-point v2 collapse at $\eta = 10^{-4}$ is attributable to stain normalization rather than to dynamic balancing; the residual GradNorm contribution at $\eta = 10^{-4}$ (with Macenko held on) requires the static 10^{-4} + Macenko control (Run 27). We therefore make no GradNorm-specific penalty claim at either learning rate and retract the earlier draft's −2.85% / −12.46% attributions: the former compared two static runs, and the latter confounds GradNorm with Macenko.

Completing the design. Three matched control runs are defined in the released pipeline to close the remaining cells: Run 27 (10^{-4}, static, Macenko on), Run 28 (10^{-3}, GradNorm, Macenko on — the originally intended Run-18 arm), and Run 29 (10^{-3}, static, Macenko on). They are pending at the time of submission; upon completion the $2 \times 2 \times 2$ decomposition is fully isolated. If Runs 28/29 reveal a GradNorm-specific penalty at $\eta = 10^{-3}$, a milder asymmetry exponent ($\alpha \leq 0.5$) or decoupled, task-specific projection heads would be the natural next step; with the current evidence that question remains open.

5 Conclusion & Recommendations

Through an exhaustive 26-experiment campaign, we demonstrated that hard-parameter-sharing U-Nets cannot achieve the inflated 88% − 99% metrics reported in recent multi-task pathology literature, and that the published claims rest on background-Dice evaluation artifacts and non-isolated data splits. Under strict isolation, the residual discrepancy on fine-grained tasks is dominated by dataset difficulty (PANDA challenge-level performance) and by the sensitivity of fine-grained tasks to the joint optimization recipe; in particular, our matched skip-connection ablation is a verified null result, and our GradNorm analysis was limited by a phase-gating defect that silently dropped one experimental arm — a defect we have now fixed and disclosed. For future computational pathology and medical imaging studies, we recommend: (1) mandatory patient/biopsy-level `GroupKFold` splitting; (2) foreground-only or zero-penalty Dice reporting on sparse-lesion datasets; (3) 95% confidence intervals on every single-run comparison; (4) validation on diverse multi-organ benchmarks such as PanNuke; and (5) auditing implementation conventions of borrowed optimization components — effective hyperparameters recorded per run, and explicit command-line overrides rather than implicit phase-level gating.

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